# 5

# The Wage Effects of Immigration: Structural Estimates

The structural approach to estimating the labor market impact of immigration has an interesting intellectual history. As noted earlier, the initial empirical studies actually imposed a structure on the technology of the local labor market, such as the translog production function, and used the resulting estimates to predict the various wage effects. The obvious problem was one of dimensionality: for a reasonably detailed specification of the skill groups that populate the labor market, the methodology quickly strained the available data. Even if there were only 10 skill groups in the workforce, for example, the researcher needed to estimate over 50 parameters to describe how a particular supply shock affected the wage of all directly competing and potentially complementary workers.

Although this approach fell out of favor in the 1990s, the past decade has witnessed renewed and increasing interest in structural methods. Keeping in mind the lesson from the early structural estimates, however, the new studies emphasize the construction of models that both reduce the dimensionality of the parameter space and build a closer link between the key theoretical concepts and the variables that are actually used in the empirical research.

## A Nested CES Model

A structural approach is required if the purpose of the analysis is not simply to estimate the "own" effect of a particular immigrant influx on the wage of directly competing native workers, but also the "cross" effects on the wage of other native workers.[1] The assumption of a specific functional form for the aggregate production function allows for the estimation of the complete set of factor price elasticities that determine how immigration affects the entire wage structure. One particularly tractable approach, introduced into the immigration literature in Borjas (2003), assumes that the aggregate technology of the labor market can be represented in terms of a three-level nested CES production function where the "primitive" labor inputs are the skill groups defined by education and experience. This section presents and estimates the nested CES framework before proceeding to discuss extensions that have been proposed in the subsequent literature.

Suppose the aggregate production function for the national labor market at time $t$ can be written as:

$$Q_t = [\lambda_{Kt} K_t^{\delta} + \lambda_{Lt} L_t^{\delta}]^{1/\delta}, \tag{5.1}$$

where $Q_t$ is output, $K_t$ is capital, $L_t$ denotes the aggregate labor input measured in efficiency units; and $\delta = 1 - 1/\sigma_{KL}$, with $\sigma_{KL}$ being the elasticity of substitution between capital and labor ($-\infty < \delta \leq 1$). The vector $\lambda$ gives (potentially) time-varying technology parameters that shift the production frontier, with $\lambda_{Kt} + \lambda_{Lt} = 1$. The total number of efficiency units $L_t$ incorporates the contributions of workers who differ in both education and experience.

The Armington aggregator giving the total number of efficiency units in the workforce is:

$$L_t = \left[ \sum_e \theta_{et} L_{et}^{\beta} \right]^{1/\beta}, \tag{5.2}$$

where $L_{et}$ gives the effective supply of workers with education $e$ at time $t$, and $\beta = 1 - 1/\sigma_E$, with $\sigma_E$ being the elasticity of substitution across these education aggregates ($-\infty < \beta \leq 1$). The $\theta_{et}$ give (potentially) time-varying technology parameters that shift the relative productivity of education groups, with $\Sigma_e \theta_{et} = 1$.

Finally, the effective supply of workers in each education group is itself given by an Armington aggregation of the contribution of similarly educated workers with different levels of work experience. In particular:

$$L_{et} = \left[ \sum_{x} \alpha_{ex} L_{ext}^{\eta} \right]^{1/\eta}, \tag{5.3}$$

where $L_{ext}$ gives the number of work hours provided by workers in education group $e$ and experience group $x$ at time $t$; and $\eta = 1 - 1/\sigma_X$, with $\sigma_X$ being the elasticity of substitution across experience classes within an education group ($-\infty < \eta \leq 1$). Equation (5.3) incorporates an important identifying assumption: the coefficients $\alpha_{ex}$ are time-invariant, with $\Sigma_x \alpha_{ex} = 1$.

The nested CES framework has two features that make it extremely useful for empirical analysis. First, it greatly reduces the dimensionality of the problem. The descriptive analysis presented in Chapter 4 classified workers into 40 different education-experience groups, so that there are 41 factors of production (including capital). Even after imposing the symmetry restrictions implied by factor demand theory, a general specification of the technology, such as the translog, would require the estimation of 861 different parameters (or $n(n+1)/2$). The nested CES drastically reduces the size of the parameter space; the technology can be summarized in terms of *three* elasticities of substitution.[2] Obviously, this simplification comes at a cost: the CES specification restricts the types of substitution that can exist among the various factors. The elasticity of substitution across experience groups takes on the same value for workers in adjacent experience categories as for workers who differ greatly in their experience; the elasticity of substitution between high school dropouts and high school graduates is the same as that between high school dropouts and college graduates; and the elasticity of substitution between capital and labor is the same for all the different types of workers as long as the labor input is measured in efficiency units.

Second, the nested CES framework is easily estimable using the national-level census data by education and experience introduced in Chapter 4. In particular, note that the marginal productivity condition implies that the wage for cell ($e$, $x$, $t$) is:

$$\begin{aligned} \log w_{ext} = {} & \log \lambda_{Lt} + (1-\delta) \log Q_t + (\delta - \beta) \log L_t + \log \theta_{et} \\ & + (\beta - \eta) \log L_{et} + \log \alpha_{ex} + (\eta - 1) \log L_{ext}. \end{aligned} \tag{5.4}$$

This marginal productivity condition can be rewritten as:

$$\log w_{ext} = \phi_t + \phi_{et} + \phi_{ex} - \frac{1}{\sigma_X} \log L_{ext}, \tag{5.5}$$

where $\phi_t = \log \lambda_{Lt} + (1-\delta) \log Q_t + (\delta - \beta) \log L_t$, and is absorbed by period fixed effects; $\phi_{et} = \log \theta_{et} + (\beta - \eta) \log L_{et}$, and is absorbed by interactions between education and period fixed effects; and $\phi_{ex} = \log \alpha_{ex}$, and is absorbed by interactions between education and experience fixed effects. In short, a regression of the log wage for a particular skill group on the log quantity of that skill group, holding constant the relevant vectors of fixed effects, identifies the elasticity of substitution across experience groups.

Because the coefficients of the education-experience fixed effects in (5.5) identify the parameters $\log \alpha_{ex}$, it is trivial to estimate the actual weights $\alpha_{ex}$ by imposing the restriction that $\Sigma_x \alpha_{ex} = 1$.[3] As implied by the Armington aggregator in equation (5.3), the estimates of $\alpha_{ex}$ and $\sigma_X$ then permit the calculation of $L_{et}$, the total number of efficiency units supplied by education group *e* at time *t*. One can then move up one level in the CES technology, and recover an additional unknown parameter. Let $\log w_{et}$ be the mean log wage paid to workers in education group *e* at time *t*. The marginal productivity condition determining the wage for this group implies that:

$$\log w_{et} = \phi_t + \log \theta_{et} - \frac{1}{\sigma_E} \log L_{et}. \tag{5.6}$$

Equation (5.6) is closely related to the Katz-Murphy (1992) relative demand function that examines how the wage differential between college and high school graduates is affected by changes in relative supplies. As in the Katz-Murphy framework, equation (5.6) introduces a technical issue with important empirical ramifications. Note that the presence of $\log \theta_{et}$ suggests that the model must include a full set of education-time fixed effects to adjust for unobservable shifts in the productivity of different education groups. However, it is impossible to include all of these interactions, because they would be perfectly collinear with $L_{et}$, the effective supply of education group *e* at time *t*. Katz and Murphy (1992) circumvented the problem by assuming that the technology shifters $\log \theta_{et}$ can be approximated by a *linear* trend that varies across education groups. Given this operational assumption, the regression model in (5.6) then yields an estimate of the elasticity of substitution $\sigma_E$. It turns out, however, that this assumption "may not be innocuous" (Card and Lemieux 2001, p. 713).

The nested CES framework raises a number of methodological issues. First, note that OLS regressions of equations (5.5) and (5.6) yield biased estimates of $\sigma_X$ and $\sigma_{\ddot{E}}$ because the supply of workers to the various skill groups is likely to be endogenous over the 50-year period spanned by the data used below. The economic question at the core of the analysis, however, suggests an instrument for the (effective) size of the workforce in each skill group: the size of the population of foreign-born workers in that group. In other words, the immigrant influx into particular skill groups provides the supply shifter required to identify the labor demand curve. This instrument would be valid if the immigrant influx into particular skill groups were independent of the relative wages offered to the various skill categories. The number of immigrants in a skill group, however, likely responds to shifts in the wage structure. Income-maximizing behavior on the part of potential immigrants would generate larger flows into those skill cells that offered relatively higher wages. This behavioral response would tend to build in a *positive* correlation between the size of the immigrant influx and wages in a skill group. The estimated regression coefficients would then understate the negative wage impact of a relative supply increase.[4]

Second, the regressions of mean log wages on log quantities in equations (5.5) and (5.6) yield only two of the three elasticities of substitution ($\sigma_X$ and $\sigma_E$) that summarize the nested CES technology. The estimation of the elasticity of substitution between capital and labor $\sigma_{KL}$ requires additional data on the aggregate capital stock or the rate of return to capital.[5] The literature typically evades this problem by assuming that the aggregate production function in equation (5.1), the top level of the nested CES system, is Cobb-Douglas (or, more generally, linear homogeneous). *This assumption is definitely not innocuous.* As I showed in Chapter 3, the Cobb-Douglas assumption predetermines the numerical value of the average wage effect of immigration. Specifically, a 10 percent immigration-induced increase in supply *must* lower the average wage by 3.0 percent in the short run and by 0.0 percent in the long run. Similarly, the more general assumption of a linear, homogeneous production function at the top level of the nesting builds in the restriction that immigration *cannot* change the average wage in the long run.

The data used to estimate the structural CES model were introduced in Chapter 4 and give the mean log weekly wage and the total hours supplied for each of the 40 education-experience groups in the national labor market between 1960 and 2010. Table 5.1 summarizes the estimated coefficients.

**Table 5.1** Estimates of (inverse) elasticities of substitution in nested CES framework

| | Men | Men and women |
|---|---|---|
| 1. Estimate of $-1/\sigma_X$ | −0.153 | −0.112 |
| | (0.060) | (0.051) |
| 2. Estimate of $-1/\sigma_E$ | | |
| a. Linear trend for demand shocks, efficiency unit counts | −0.257 | −0.252 |
| | (0.137) | (0.111) |
| b. Linear trend for demand shocks, actual hours counts | −0.234 | −0.233 |
| | (0.115) | (0.095) |
| c. Linear trend and post-1990 spline, efficiency unit counts | −0.215 | −0.222 |
| | (0.097) | (0.072) |
| d. Linear trend and post-1990 spline, actual hours counts | −0.200 | −0.208 |
| | (0.084) | (0.063) |
| e. Quadratic trend, efficiency unit counts | −0.297 | −0.223 |
| | (0.353) | (0.274) |
| f. Cubic trend, efficiency unit counts | −0.234 | −0.218 |
| | (0.245) | (0.244) |
| g. Log trend, efficiency unit counts | 0.099 | 0.061 |
| | (0.067) | (0.060) |

*Sources:* U.S. Census PUMS, 1960–2010; see Appendix B for details.

*Notes:* The analysis is restricted to workers aged 18–64, who are classified into five education groups and eight experience groups. Robust standard errors are reported in parentheses and are clustered at the education-experience level in row 1. The dependent variable is the mean log weekly wage of natives in an education-experience-year cell in row 1 (men in the first data column; men and women in the second); and the mean log weekly wage of natives in an education-year cell in rows 2a–2g. The independent variable is the log size of the workforce in that cell, either in "efficiency units" or "actual hours" counts, The mean log weekly wage is calculated in the sample of natives employed in the wage-and-salary sector who are not enrolled in school; the size of the workforce is calculated over all workers (including men and women). All regressions are estimated using instrumental variables, where the instrument is the log of the number of foreign-born persons (both men and women) in the cell. The regressions in row 1 include vectors of year, education-year, and education-experience fixed effects. The regressions in rows 2a–2g include education-specific trends; the trend variable is initialized at 1 in 1960. The "post-1990" spline regressions add interactions between the education fixed effects and an indicator variable set to one in the 1990–2010 censuses. The regressions are weighted by the number of observations used to calculate the dependent variable. The regressions reported in row 1 have 240 observations; the regressions in rows 2a–2g have 30 observations.

The first row of the table reports that the IV estimation of equation (5.5) yields an estimate of $-1/\sigma_X$ equal to −0.153 (0.060). The implied estimate of the elasticity of substitution across experience groups is 6.7.

Given the estimate of $\sigma_X$ and of the weights $\alpha_{ex}$, it is then possible to use the aggregator in equation (5.3) to calculate $L_{et}$. This calculation, in turn, allows the estimation of equation (5.6), which yields an estimate of $\sigma_E$, the elasticity of substitution across education groups. As noted above, the estimation of this equation requires that some assumption be made about the differential *and unobserved* trends in productivity across education groups (that is, the time variation in $\theta_{et}$). Row 2a of Table 5.1 shows that the estimate of $-1/\sigma_E$ is −0.257 (0.137) if we maintain the original Katz-Murphy assumption of a linear trend that differs across education groups. The implied elasticity of substitution across education groups is 3.9. Row 2b reestimates the regression by using the total number of hours worked by workers in each education group, rather than the Armington aggregation that relies on $\sigma_X$ and the estimated $\alpha_{ex}$. The implied elasticity of substitution across education groups is robust to this specification change.

Autor et al. (2008) argue that the original Katz-Murphy linear trend assumption fails to capture some of what happened to the U.S. wage structure in the 1990s. Specifically, they document that the growth in the relative demand for skilled workers slowed in the 1990s. To capture this break in the linear trend, I included additional interactions in equation (5.6) that allow for differential shifts in the education-specific trends since 1990. Specifically, row 2c estimates a regression that allows for a "spline" by including not only interactions between the education fixed effects and a linear trend, but also interactions between the education fixed effects and an indicator that "turns on" in 1990.[6] Row 2c shows that the estimate of $-1/\sigma_E$ with this expanded spline specification is −0.215 (0.097), implying that the elasticity of substitution across education groups is around 5.0. I will use this implied value of the elasticity of substitution across education groups in the simulations reported in the next section.

It is important to emphasize, however, that the estimate of the inverse elasticity becomes much more imprecise (and can even change sign) under alternative trend assumptions. Row 2e, for example, assumes that the unobserved demand shifts across education groups follow a quadratic trend, while row 2f assumes a cubic trend. Both of these alternative assumptions lead to a dramatic increase in the standard error so that the elasticity of

substitution $\sigma_E$ is no longer statistically significant. Finally, the assumption of a log trend in row 2g leads to a breakdown of the empirical approach: the implied elasticity of substitution has the wrong sign.

## Simulating the Wage Effects of Immigration

The nested CES model provides a simple method for calculating the predicted wage effects of immigration. Consider, in particular, the impact of a "generalized" supply shift on the wage of a skill group with education *e* and labor market experience *x*. By differentiating the marginal productivity condition for this group, it is easy to show that:

$$d\log w_{ex} = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} + \left(\frac{1}{\sigma_X} - \frac{1}{\sigma_E}\right) m_e - \frac{1}{\sigma_X} m_{ex}, \tag{5.7}$$

where $m_{ex} = d \log L_{ex}$, the percent shift in the supply of skill group (*e, x*); $m_e = d \log L_e$; and $\bar{m} = d \log L$.

Equation (5.7) has a number of interesting properties. First, as emphasized in Chapter 3, it is not necessary to estimate any regressions to calculate the magnitudes of the immigration-induced supply shifts $m_e$ and $\bar{m}$. Specifically, $m_e = \sum_x s_{ex} m_{ex} / s_e$, and $\bar{m} = \sum_e s_e m_e / s_L$. In other words, the supply shift occurring at any level of the nested CES framework is simply a weighted average of the supply shifts observed at a lower level of aggregation—where the weights are given by income shares.[7]

Second, one important "deliverable" from this type of empirical exercise is the prediction of the wage impact of immigration across the various education groups. It is easy to calculate the mean wage effect for the various education groups by taking a weighted average of the wage effects in equation (5.7) across the experience groups for a particular education group *e*. The weight implied by factor demand theory would be the relative income share accruing to each of the experience groups in education group *e*, or ($s_{ex}/s_e$). This weighted average yields:[8]

$$d\log w_e = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} - \frac{1}{\sigma_E} m_e, \tag{5.8}$$

Equation (5.8) again illustrates a remarkable property of the nested CES: the impact of an immigration-induced supply shift on the average wage of education group *e* is independent of any elasticities of substitution that enter the model *below* the education level in the CES nesting (in the current context, it is independent of $\sigma_X$). The predicted wage effect on education group *e* depends only on the elasticity of substitution between labor and capital, and on the elasticity of substitution across education groups.[9]

Finally, recall that the generic model in the literature does not attempt to estimate the elasticity of substitution between labor and capital, $\sigma_{KL}$. Instead, the numerical value of this elasticity is built in by the assumption that the aggregate production function in the nested CES framework is Cobb-Douglas, or $\sigma_{KL} = 1$. The impact of an immigration-induced supply shift on the mean wage in the labor market can again be obtained by taking a weighted average of the mean wage effect across education groups using relative income shares (or $s_e/s_L$) as the weights. The average wage effect is:

$$d\log\bar{w} = \frac{s_K}{\sigma_{KL}}(d\log K - \bar{m}) = s_K(d\log K - \bar{m}). \tag{5.9}$$

As expected, equation (5.9) implies that the mean wage effect is zero in the long run (when $d \log K = \bar{m}$ ) and would simply equal $-s_K$ times $\bar{m}$ in the short run, when $d \log K = 0$. Again, note the property that the aggregate wage effect does not depend directly on any of the elasticities of substitution that enter the nested CES framework below the level of the aggregate labor input (specifically, it does not depend on $\sigma_E$ and $\sigma_X$). Put bluntly, the voluminous amount of empirical work that relates mean wages in specific skill groups to group-specific quantities *provides no information whatsoever* to help calculate the mean wage effect of immigration on the host country's labor market. The mean wage effect reported in these types of simulations is the mechanical result of the assumption that the aggregate production function is Cobb-Douglas (or linear homogeneous) and has nothing to do with the underlying data.

Table 5.2 summarizes the results of the simulation. Specifically, the table uses the estimated elasticities of substitution reported in rows 1 and 2c of Table 5.1 (that is, $\sigma_X = 6.7$ and $\sigma_E = 5.0$) to calculate the percent wage impact resulting from the generalized supply shift induced by immigration between 1990 and 2010. The first row of Table 5.2 shows that the supply shift was particularly large for workers at the bottom and top of the

**Table 5.2** Simulated wage impact of 1990–2010 immigrant influx on the preexisting workforce

| | High school dropouts | High school graduates | Some college | College graduates | Post-college | All workers |
|---|---|---|---|---|---|---|
| Percent supply shift | 25.9 | 8.4 | 6.1 | 10.9 | 15.0 | 10.6 |
| Percent wage effects | | | | | | |
| Short run | −6.2 | −2.7 | −2.3 | −3.2 | −4.1 | −3.2 |
| Long run | −3.1 | 0.4 | 0.9 | −0.1 | −0.9 | 0.0 |

*Source:* Elasticities estimated in Table 5.1, male sample, rows 1 and 2c.

*Notes*: The simulations use the nested CES framework, set $\sigma_X = 6.7$, $\sigma_E = 5.0$, and assume that the aggregate production function is Cobb-Douglas, implying $\sigma_{KL} = 1.0$. The preexisting workforce includes both natives and immigrants residing in the United States in 1990. The short-run simulation assumes that the capital stock is constant; the long-run simulation assumes that the rate of return to capital is constant. Relative wage effects may differ between the short and long runs due to rounding error.

education distribution.[10] Immigration increased the effective number of work hours of high school dropouts by 25.9, and those of workers with more than a college degree by 15.0 percent. In contrast, immigration increased the number of hours supplied by workers with 12 to 15 years of school by 6 to 8 percent. Overall, immigration increased effective supply by 10.6 percent during the two-decade period.

Because of the U-shaped nature of the supply shift, the simulation necessarily shows that immigration most affected the wage of workers at the two ends of the education distribution. The estimated elasticity of substitution across education groups implies that the large immigration-induced supply shift experienced by high school dropouts reduced the wage of this group by 6.2 percent in the short run and 3.1 percent in the long run. Similarly, the wage effect for the most highly skilled workers (those with more than a college degree) were −4.1 percent in the short run and −0.9 percent in the long run.[11]

If we take the weighted average of the wage effects across education groups, we find that the average wage of a preexisting worker fell by 3.2 percent in the short run and 0.0 percent in the long run. It is important to reemphasize, however, that these aggregate effects have nothing to do with the underlying data. They are the mechanical predictions of the assumption that the aggregate production function is Cobb-Douglas (so, for example, the short-run average wage effect of 3.2 percent is simply the product of capital's share of income, 0.3, times the aggregate supply shift, 10.6 percent).

The mechanical nature of these aggregate wage effects suggests that we should view the absolute numerical values provided by a simulation that relies on a nested CES/Cobb-Douglas framework with *more* than the usual skepticism. The Cobb-Douglas assumption implies that the average wage effect in the short run *must be* −3.2 percent. Therefore, some education groups must have experienced a wage loss that is somewhat larger than 3.2 percent, while other education groups must have experienced a somewhat smaller wage loss. Despite its simplicity and elegance, the nested CES framework carries a great deal of baggage as well, and this baggage suggests that the only valuable results that come out of the simulation deal with the impact of immigration on *relative* wages. Although obvious, this fact has not been sufficiently emphasized or appreciated in the existing literature.

In addition, it is worth stressing that the distinction between the short run and the long run used in the simulations is a purely theoretical construct. As will be noted in Chapter 6, remarkably little is known about how labor markets actually adjust to immigration and about the speed of that adjustment. The available evidence simply does not provide a useful guide to help us discern which of the two sets of simulation results seems more representative of what happens in real-world labor markets. Moreover, the only results that are data-driven in these simulations measure the impact of immigration on relative wages. However, as equation (5.8) shows, the algebra of the nested CES *forces* these relative wage effects to be the same in both the short and the long run. In the end, therefore, the distinction between short run and long run is a red herring. The simulated wage effects can be made to look "small" or "large" by choosing a particular time horizon, but this choice does not offer any real-world informational content.

## Within-Group Complementarity between Immigrants and Natives

Both the descriptive empirical analysis reported in Chapter 4 and the structural analysis presented in the previous section assumed that immigrant and native workers belonging to a particular skill group $(e, x)$ are perfect substitutes.[12] The question of whether there is *within-group* imperfect substitution—giving rise to potential complementarities between similarly skilled immigrants and natives—was first examined in the context of a structural framework by Jaeger (1996), who concluded that the two groups were indeed perfect substitutes. More recently, however, Ottaviano

and Peri (2006, 2012) revisited this question and report some evidence for within-group complementaries. In fact, in the original version of their work, they concluded that the complementarities were sufficiently strong that an immigration-induced supply shock would increase the wage of almost all native workers.

The extent of within-group complementarities can be easily determined in the nested CES framework by adding a fourth level to the production technology. In particular, let $N_{ext}$ and $M_{ext}$ be the total number of work hours supplied by natives and immigrants, respectively, with education *e,* experience *x,* at time *t*. The Armington aggregator describing the interactions between these two groups is given by:

$$L_{ext} = [\varphi_{ext} N_{ext}^{\gamma} + (1-\varphi_{ext}) M_{ext}^{\gamma}]^{1/\gamma}, \tag{5.10}$$

where $\gamma = 1 - 1/\sigma_{MN}$, with $\sigma_{MN}$ being the elasticity of substitution between equally skilled foreign- and native-born workers (and $-\infty < \gamma \leq 1$).

Note that the type of complementarity measured by $\sigma_{MN}$ is quite specific and narrow. It does not describe how low-skill immigrants and high-skill natives may interact in the production process—which is the type of complementarity that could easily be presumed to be empirically relevant in the context of immigration to many developed countries. Instead, it describes the complementarities that arise when a 30-year-old foreign-born high school dropout interacts with a 30-year-old native-born high school dropout.[13]

Equating wages to the value of marginal product of labor yields the estimable relative labor demand function:

$$\log \frac{w_{ext}^{M}}{w_{ext}^{N}} = \varphi_{ext}^{\star} - \frac{1}{\sigma_{MN}} \log\left(\frac{M_{ext}}{N_{ext}}\right), \tag{5.11}$$

where $w_{ext}^{M}$ and $w_{ext}^{N}$ give the wage of immigrant and native workers in cell (*e, x, t*), respectively; and $\varphi_{ext}^{\star} = \log\,[(1-\varphi_{ext}) / \varphi_{ext}]$, a (potentially) time- and skill-varying relative demand shifter that will presumably need to be approximated by a vector of observed variables.

The first row of Table 5.3 reports the inverse elasticity of substitution estimated in the Ottaviano and Peri (2012) study. The inverse elasticity of substitution estimated using the simplest specification is −0.053 (0.008), implying that $\sigma_{MN}$ is around 20.[14] It is important to emphasize that this

**Table 5.3** Estimates of inverse elasticity of substitution between equally skilled immigrants and natives ($-1/\sigma_{MN}$)

| | Specification | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| A. Log of the mean wage (Ottaviano-Peri, 2012) | | | | |
| Men | −0.053 | — | −0.033 | — |
| | (0.008) | | (0.013) | |
| Men and women | −0.032 | — | −0.024 | — |
| | (0.008) | | (0.015) | |
| B. Mean of the log wage (Borjas et al. 2012) | | | | |
| Men | −0.008 | 0.008 | 0.008 | 0.009 |
| | (0.017) | (0.021) | (0.013) | (0.034) |
| Men and women | −0.002 | 0.004 | 0.001 | −0.034 |
| | (0.015) | (0.018) | (0.012) | (0.036) |
| C. Mean of the log wage, updated to 2010 | | | | |
| Men | −0.018 | 0.005 | −0.003 | 0.029 |
| | (0.007) | (0.009) | (0.009) | (0.019) |
| Men and women | −0.036 | −0.011 | −0.017 | 0.016 |
| | (0.006) | (0.006) | (0.015) | (0.017) |
| Fixed effects included | | | | |
| Year | No | Yes | Yes | Yes |
| Education × experience | No | No | Yes | Yes |
| Education × year | No | No | No | Yes |
| Experience × year | No | No | No | Yes |

*Sources:* Adapted from Borjas et al. (2012, p. 202). The inverse elasticities reported in Panel A are drawn from Ottaviano and Peri (2012, p. 171, columns 1 and 2).

*Notes:* Robust standard errors reported in parentheses. The unit of observation is an education-experience-year cell. The dependent variable is the difference in the log weekly wage between immigrants and natives in the cell, and the independent variable is the corresponding difference in the log total hours supplied. Panel A defines the log wage of a group as the log of the mean wage across workers in that group; Panels B and C define it as the mean of the log wage across workers. The regressions reported in Panel A use total employment in the skill group as the weight; in all other panels, the weight is the inverse of the sampling variance of the dependent variable. The regressions in Panel C use the same data employed in the regressions reported in row 1 of Table 5.1 (with the mean log wage for an education-experience group being calculated separately for natives and immigrants).

"preferred" estimate is based on a regression model that treats $\varphi^{*}_{ext}$ in equation (5.11) as a constant and simply regresses the log of relative wages on the log of relative quantities.

It turns out, however, that an estimated elasticity of 20 exaggerates the within-group complementarities that are actually revealed by the census data. Borjas et al. (2012) replicated the Ottaviano-Peri study and found that the estimate of $\sigma_{MN}$ would have been much larger had the regression analysis used modeling assumptions that are widely accepted in the labor economics literature. Specifically, Ottaviano and Peri define the value of marginal product for a skill group as the *log of the mean wage* for a particular cell. The standard approach in the literature is to use the *mean of the log wage*.[15] By Jensen's inequality, the log mean wage and the mean log wage are not the same. The mean log wage corresponds to the geometric mean wage, which has the interpretation of a marginal product index in which weights are each worker's share of total hours worked. The log mean wage does not have such an economic interpretation. It turns out that this seemingly trivial definitional change has a major impact on the estimates.

Panel B of Table 5.3 reports the estimated inverse elasticity resulting from a regression that uses the mean log wage. The regression coefficient is now −0.008 (0.017). The coefficient is no longer statistically significant and has an implied substitution elasticity of 125. Finally, Panel C reports the coefficients from regressions that use the updated 1960–2010 census data. The estimate of the inverse elasticity of substitution is −0.018, implying an elasticity of over 50.

Further, note that the coefficients reported using the simplest regression specification ignore the possibility of time-skill variation in the intercept $\varphi^{*}_{ext}$ in equation (5.11). Because the dependent variable in this regression is the difference in immigrant-native log wages, factors that affect immigrant and native labor demand equally are automatically removed from the equation. Remaining factors that affect immigrant-native relative wages include differences in the composition of immigrants and natives within skill groups and differential shifts in demand.[16] One solution is to control for skill-group and period fixed effects. The inclusion of these fixed effects consistently produces estimated inverse elasticities that are both numerically and statistically equal to zero. Put simply, the available evidence suggests that within-group complementarities between foreign- and native-born workers are not an important factor in an assessment of the labor market impact of immigration in the United States.[17]

It is instructive to illustrate the role played by potential within-group complementarities in simulations of the labor market impact of immigration. By differentiating the marginal productivity conditions giving the wage of foreign- and native-born workers in a particular skill group, it is easy to show that the wage impact of an immigration-induced supply shift is given by:

$$d\log w_{ex}^{M} = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} + \left(\frac{1}{\sigma_X} - \frac{1}{\sigma_E}\right)m_e + \left(\frac{1}{\sigma_{MN}} - \frac{1}{\sigma_X}\right)m_{ex} - \frac{1}{\sigma_{MN}} d\log M_{ex}, \tag{5.12}$$

$$d\log w_{ex}^{N} = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} + \left(\frac{1}{\sigma_X} - \frac{1}{\sigma_E}\right)m_e + \left(\frac{1}{\sigma_{MN}} - \frac{1}{\sigma_X}\right)m_{ex}. \tag{5.13}$$

Equations (5.12) and (5.13) imply that the impact of immigration on the wage differential between equally skilled foreign- and native-born workers, or $d\log(w_{ex}^{M} / w_{ex}^{N})$, depends on the product of the inverse elasticity $\sigma_{MN}$ and $d\log M_{ex}$, the percent change in the number of work hours supplied by foreign-born workers. If the elasticity of substitution $\sigma_{MN}$ were small, the within-group complementarities would inevitably have a substantial impact on the wage gap between comparably skilled foreign- and native-born workers.

However, the nested CES framework also implies that the value of $\sigma_{MN}$ plays no role in determining the impact of immigration on the average wage of the preexisting workforce for any particular skill group—where the preexisting workforce includes both the native-born and the preexisting foreign-born workers. In other words, the average wage effect on all preexisting workers in a particular education group *must be identical* to that estimated in the previous section, regardless of the value of $\sigma_{MN}$.

Table 5.4 summarizes the results of the simulation, using two alternative values for the elasticity: 20 and infinity. The elasticity of 20 is the Ottaviano-Peri preferred estimate; and an elasticity of infinity is the estimate that, in fact, seems most consistent with the 1960–2010 census data. To simplify the exposition, the table reports only the long-run wage impacts. It

**Table 5.4** Simulated long-run wage impact of 1990–2010 immigrant influx, allowing for within-group complementarities

| | High school dropouts | High school graduates | Some college | College graduates | Post-college | All workers |
|---|---|---|---|---|---|---|
| Percent wage effects, $\sigma_{MN} = 20.0$ | | | | | | |
| Native-born | −1.7 | 0.9 | 1.2 | 0.5 | −0.1 | 0.6 |
| Foreign-born | −5.3 | −3.4 | −2.7 | −4.9 | −5.3 | −4.4 |
| All workers | −3.1 | 0.4 | 0.9 | −0.1 | −0.9 | 0.0 |
| Percent wage effects, $\sigma_{MN} = \infty$ | | | | | | |
| All workers | −3.1 | 0.4 | 0.9 | −0.1 | −0.9 | 0.0 |

*Source:* Elasticities estimated in the male sample of Table 5.1, rows 1 and 2c; Table 5.3, Panel A, data column 1.

*Notes:* The simulations use the nested CES framework, set $\sigma_X = 6.7$, $\sigma_E = 5.0$, and assume that the aggregate production function is Cobb-Douglas, implying $\sigma_{KL} = 1.0$. The long-run simulation assumes that the rate of return to capital is constant; the short-run wage effects can be obtained by adding −3.2 to any statistic reported in the table.

should be obvious that the short-run wage impact for any specific group can be obtained by subtracting 3.2 percent from any statistic in the table.

The simulation, of course, shows that the wage effect on the *average* pre-existing worker in each education group is the same regardless of the value of the elasticity. If the elasticity is less than infinity, however, there is a distributional effect within each education group, with immigrants being relatively harder hit by the supply shock. Note, however, that the wage of native-born high school dropouts falls (in the long run) by between 2 and 3 percent regardless of the value of the elasticity. In other words, even an elasticity $\sigma_{MN}$ "as low" as 20 has roughly the same operational consequences as the assumption that equally skilled immigrants and natives are perfect substitutes.

## Are "High School Equivalents" Equivalent?

The aggregation of workers into a manageable number of skill groups is a crucial step in any empirical analysis of the labor market impact of immigration. In the United States, immigration has disproportionately increased the size of specific education groups, such as high school dropouts

and workers with post-college degrees. Hence, the literature has focused on estimating the impact of immigration on those particular groups. In contrast, much of the wage structure literature found it convenient to discuss trends in the returns to skills by examining the wage gap between two broadly defined education classifications, "high school equivalents" (defined as an aggregation of high school dropouts and high school graduates), and "college equivalents" (defined as an aggregation of workers who have more than a high school diploma). A number of recent studies argue that the high school equivalents–college equivalents classification should also be adopted in the immigration literature.[18]

Of course, whether such an adoption is warranted is an empirical question, depending partly on the value of the elasticity of substitution between high school dropouts and high school graduates.[19] Equation (5.2) defined the Armington aggregator giving the total number of efficiency units supplied in the labor market as $L_t = [\sum_e \theta_{et} L_{et}^{\beta}]^{1/\beta}$, where the index $e$ distinguishes the five education groups ($e = 1, \ldots, 5$). The notion that high school dropouts and high school graduates may be easily substitutable can be captured by specifying yet another level in the nested CES framework. In particular, the aggregator in equation (5.2) is replaced by the two-equation system:

$$\begin{aligned} L_t &= \left[\theta_{HS,t} L_{HS,t}^{\beta} + \theta_{3t} L_{3t}^{\beta} + \theta_{4t} L_{4t}^{\beta} + \theta_{5t} L_{5t}^{\beta}\right]^{1/\beta}, \\ L_{HS,t} &= \left[\mu_t L_{1t}^{\rho} + (1-\mu_t) L_{2t}^{\rho}\right]^{1/\rho}, \end{aligned} \tag{5.14}$$

where $L_{HS,t}$ gives the efficiency units provided by high school equivalents; and $\rho = 1 - 1/\sigma_{HS}$, with $\sigma_{HS}$ being the elasticity of substitution between high school dropouts and high school graduates ($-\infty < \rho \leq 1$).[20] The marginal productivity conditions then imply that the relative demand function for high school graduates/dropouts is given by:

$$\ln\left(\frac{w_{2t}}{w_{1t}}\right) = \mu_t^* - \frac{1}{\sigma_{HS}} \ln\left(\frac{L_{2t}}{L_{1t}}\right), \tag{5.15}$$

where $\mu_t^* = \log[(1-\mu_t)/\mu_t]$. Note that national-level data provide only one observation per year to estimate $\sigma_{HS}$. This limited variation implies that one cannot control for the demand shifter $\mu_t^*$ by simply including a vector of year fixed effects; additional assumptions about the nature of the trend are required.

Most studies in the wage structure literature use the annual Current Population Surveys (CPS) to estimate elasticities of substitution across (aggregated) education groups. The early studies in this literature assumed that the demand shifter—in the context of the wage gap between high school equivalents and college equivalents—could be approximated by a linear trend that differed between the two groups. More recent studies, however, argue that that there was a slowdown in demand growth for more skilled workers after 1990. Goldin and Katz (2010) adopt this framework, and add a post-1992 spline to the linear trend, to specifically estimate $\sigma_{HS}$, the elasticity of substitution between high school dropouts and high school graduates.

Borjas et al. (2012) used the Goldin-Katz CPS data on relative wages and employment for the 1963–2005 period to evaluate the robustness of estimates of $\sigma_{HS}$. Table 5.5 summarizes the results. The first coefficient reported in the table gives the elasticity $\sigma_{HS}$ estimated using the "preferred" spline specification.[21] The estimated coefficient is −0.135 (0.023), rejecting the hypothesis that the two groups are perfect substitutes. The implied elasticity $\sigma_{HS}$ is 7.4.

The other rows in Table 5.5 use alternatives to the spline approximation for the demand shifter. It is evident that the estimated $\sigma_{HS}$ is extremely sensitive to how the regression controls for the unobserved variation in relative demand. Row 2, for example, uses the original linear trend assumption. The estimated elasticity is now equal to 43.5, suggesting that high school dropouts and high school graduates are indeed near-perfect substitutes. The use of a quadratic trend leads to a rejection of the CES framework. The implied elasticity of substitution is statistically significant and sizable, but it has the wrong sign. The use of a cubic trend leads to further instability.

Card (2009) proposed an approach that would seem to avoid the pitfalls inherent in making untestable assumptions about the underlying trends in relative demand. He exploits variation in relative wages and quantities across local labor markets to estimate $\sigma_{HS}$. In other words, the value of the elasticity of substitution is inferred from a spatial correlation between relative wages and relative quantities. The comparison of local labor markets leads to the conclusion that the two groups are perfect substitutes. The MSA-level regression reported in Table 5.5 replicates Card's results by using the census-ACS data and defining a cell in terms of metropolitan area *r*, education *e*, and time *t*. Row 5, which includes period fixed effects, shows that the inverse elasticity is −0.024 (0.016), so that the hypothesis that high school dropouts and high school graduates are perfect substitutes cannot be rejected.

**Table 5.5** Estimates of inverse elasticity of substitution between high school dropouts and high school graduates ($-1/\sigma_{HS}$)

| | | Census | |
|---|---|---|---|
| Specification | 1. CPS | 2. MSA level | 3. State level |
| 1. Linear trend, post-1992 spline | −0.135<br>(0.027) | — | — |
| 2. Linear trend only | −0.023<br>(0.012) | — | — |
| 3. Quadratic trend | 0.101<br>(0.045) | — | — |
| 4. Cubic trend | −0.047<br>(0.012) | — | — |
| 5. Period fixed effects | — | −0.024<br>(0.016) | −0.069<br>(0.025) |
| 6. State/MSA effects × linear trend | — | −0.037<br>(0.026) | −0.119<br>(0.011) |
| 7. State/MSA effects × post-1992 spline | — | 0.141<br>(0.074) | −0.152<br>(0.028) |
| Includes state or MSA fixed effects | — | Yes | Yes |
| Number of observations | 43 | 1071 | 306 |

*Source:* Borjas et al. (2012, p. 203).

*Notes:* Robust standard errors are reported in parentheses, and are clustered at the MSA or state level in the last two columns. The unit of observation is a calendar year in rows 1–4, and a region-year cell in rows 5–7. The dependent variable gives the difference in the mean log wage between high school graduates and high school dropouts in a particular labor market at a point in time. The independent variable gives the corresponding difference in the log total hours supplied. The CPS regressions use the 1963–2005 CPS time series created by Goldin and Katz (2010, p. 306). The regression coefficients reported in the last two columns use data extracts constructed from decennial census and ACS samples. The post-1992 spline uses the definition in Goldin-Katz and is an additional linear trend variable initialized at 1 in 1993. The regressions using the CPS data are unweighted; all other regressions are weighted by the inverse of the variance of the dependent variable.

The benefit of the spatial approach is that it superficially seems to eliminate the need to make arbitrary assumptions about the unobserved demand shifter. Because there are now many observations in each period (for different localities), the regression can include period fixed effects. However, there are likely to be geographic differences in the evolution of relative demand for high school dropouts and high school graduates. Thus, the need arises to make ancillary assumptions about the *MSA-specific* trends in relative demand. Rows 6 and 7 of the table illustrate what happens to the estimated inverse elasticity when the MSA fixed effects are interacted either with a linear trend or with the spline specification. The estimated coefficients are unstable and, in the case of the spline specification, reject the CES framework.

Finally, Table 5.5 also reports the estimated elasticity when the regional labor market is defined at the state level rather than for a metropolitan area. As emphasized in Chapter 4, spatial correlations between wages and immigration are sensitive to the geographic definition of the labor market. As it turns out, so is the estimated elasticity of substitution $\sigma_{HS}$. The basic state-level regression reported in row 5 implies an elasticity of 14.5, rejecting the conjecture that high school dropouts and high school graduates are perfect substitutes. Interestingly, the estimated elasticity $\sigma_{HS}$ becomes numerically larger the smaller the geographic area that defines the labor market: it is 7.4 in the national CPS data (using the spline specification), 14.5 at the state level, and 41.7 at the MSA level. This finding mirrors the finding that wages and supply shifts are more closely linked the larger the size of the geographic labor market.

In short, the available evidence on the elasticity of substitution between high school dropouts and high school graduates is not robust to the assumptions made about the time path of *unobserved* shocks to relative demand. Different assumptions about these demand shifts yield different conclusions about whether high school equivalents are, in fact, equivalent. The sensitivity of the results suggests that the nested CES framework does not provide a particularly useful approach for analyzing the substitutability of labor between these two skill groups.

Nevertheless, it is useful to conduct one last simulation exercise to get a sense of the substantive importance of the value of the elasticity of substitution between high school dropouts and high school graduates. The CES aggregation of education groups in (5.14) implies that equations (5.12) and (5.13) still give the predicted wage effects for foreign- and native-born

workers with more than a high school diploma. For workers who are either high school dropouts or high school graduates, the predicted wage impact is now given by:

$$d\log w_{ex}^{M} = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} + \left(\frac{1}{\sigma_{HS}} - \frac{1}{\sigma_E}\right) m_{HS} + \left(\frac{1}{\sigma_X} - \frac{1}{\sigma_{HS}}\right) m_e + \left(\frac{1}{\sigma_{MN}} - \frac{1}{\sigma_X}\right) m_{ex} - \frac{1}{\sigma_{MN}} d\log M_{ex}, \tag{5.16}$$

$$d\log w_{ex}^{N} = \frac{s_K}{\sigma_{KL}} d\log K + \left(\frac{1}{\sigma_E} - \frac{s_K}{\sigma_{KL}}\right)\bar{m} + \left(\frac{1}{\sigma_{HS}} - \frac{1}{\sigma_E}\right) m_{HS} + \left(\frac{1}{\sigma_X} - \frac{1}{\sigma_{HS}}\right) m_e + \left(\frac{1}{\sigma_{MN}} - \frac{1}{\sigma_X}\right) m_{ex}. \tag{5.17}$$

where $m_{HS} = d \log L_{HS} = (s_1\ d \log L_1 + s_2\ d \log L_2)/s_{HS}$; and $s_{HS}$ is the income share accruing to workers who are either high school dropouts or high school graduates. It is again important to emphasize that the extent of substitution between high school dropouts and high school graduates does not affect the mean wage effect of immigration. The mean effect has been predetermined numerically by the Cobb-Douglas assumption.

Table 5.6 summarizes the result of the "generalized" long-run simulation using two alternative values for the elasticity $\sigma_{HS}$ : 7.4 (the value implied by the Goldin-Katz CPS data), and infinity (the assumption that the two groups are perfect substitutes). The table again uses two alternative values (20 and infinity) for $\sigma_{MN}$.

The conjecture that high school dropouts and high school graduates are perfect substitutes has several interesting implications. Most obviously, immigration has a much weaker impact on the wage of low-skill workers. For example, the long-run wage impact on high school dropouts is essentially zero if $\sigma_{HS}$ is infinity, but –2.1 percent if $\sigma_{HS}$ is around 7. The weaker impact implied by the assumption that high school dropouts and high school graduates are perfect substitutes arises mechanically. As Borjas et al. (1997) noted, because immigration disproportionately increased the number of high school dropouts in the United States, the identification of high school dropouts as a unique skill group implies that this group experienced a very large supply shock. By pooling all workers with 12 or fewer years of education into "high school equivalents," the relative size of the

**Table 5.6** Generalized simulation of the long-run wage impact of the 1990–2010 immigrant influx

| | | High school dropouts | High school graduates | Some college | College graduates | Post-college | All workers |
|---|---|---|---|---|---|---|---|
| Percent wage effects, $\sigma_{MN}=20.0$ | | | | | | | |
| $\sigma_{HS}=7.4$ | Native born | −0.8 | 0.6 | 1.2 | 0.5 | −0.1 | 0.6 |
| | Foreign born | −4.3 | −3.6 | −2.7 | −4.9 | −5.3 | −4.2 |
| | All workers | −2.1 | 0.2 | 0.9 | −0.1 | −0.9 | 0.0 |
| $\sigma_{HS}=\infty$ | Native born | 1.1 | 0.2 | 1.2 | 0.5 | −0.1 | 0.5 |
| | Foreign born | −2.4 | −4.1 | −2.7 | −4.9 | −5.3 | −4.0 |
| | All workers | −0.2 | −0.2 | 0.9 | −0.1 | −0.9 | 0.0 |
| Percent wage effects, $\sigma_{MN}=\infty$ | | | | | | | |
| $\sigma_{HS}=7.4$ | All workers | −2.1 | 0.2 | 0.9 | −0.1 | −0.9 | 0.0 |
| $\sigma_{HS}=\infty$ | All workers | −0.2 | −0.2 | 0.9 | −0.1 | −0.9 | 0.0 |

*Source:* Elasticities estimated in the male sample of Table 5.1, rows 1 and 2c; Table 5.3, Panel A, data column 1; and Table 5.5, row 1.

*Notes:* The simulations use the nested CES framework, set $\sigma_X=6.7$, $\sigma_E=5.0$, and assume that the aggregate production function is Cobb-Douglas, implying $\sigma_{KL}=1.0$. The long-run simulation assumes that the rate of return to capital is constant; the short-run wage effects can be obtained by adding −3.2 to any statistic reported in the table.

immigration-induced supply shock on the low-skill workforce becomes much smaller, and the wage impact on low-skill workers gets diluted.

Second, the assumption that high school dropouts and high school graduates are perfect substitutes implies that immigration had little distributional impact. For example, if the elasticity $\sigma_{MN}$ were infinity, most education groups suffered a negligible (long-run) wage effect. The seemingly unbalanced skill composition of recent immigration does not translate into relative wage effects because the aggregation of high school dropouts and graduates into a single group leads to effective supply shifts that are roughly of the same magnitude for *every* education group.

Finally, the comparison of Tables 5.2 and 5.6 suggests that the traditional separation of the low-skill workforce into high school dropouts and high school graduates leads to predicted relative wage effects that are largely consistent with the value of $\sigma_{HS}$ implied by the Goldin-Katz CPS

data (that is, $\sigma_{HS} = 7.4$). In Table 5.2, for example, immigration is predicted to lower the wage of high school dropouts *relative* to college graduates by 3.0 percentage points. In Table 5.6, the relative wage of high school dropouts falls by 2.0 percent.

It is best to conclude by emphasizing an obvious, but important, lesson. The simulation illustrates the limits to what the nested CES framework can teach us about the labor market impact of immigration. Aggregating or disaggregating education groups can make the estimated wage effects as small or as large as one would like. Yet, there is no convincing evidence on how best to pool.

## Descriptive versus Structural Approaches

So what is the "take-away" message from the descriptive and structural approaches to a core question in immigration economics? Specifically, what *is* the effect of an immigration-induced supply shock on the wage structure?

As the discussion in this chapter shows, the nested CES structural approach seems far too sensitive to the imposition of unverifiable (but necessary) assumptions to be of much use in giving a robust and convincing answer. Moreover, the size of the predicted effect on the average wage is driven *entirely* by functional form assumptions about the production technology, rather than estimated from data. At the same time, it is clear that a comprehensive answer requires that we take into account all the cross-effects that occur as immigrants in one skill group influence the productivity of workers in other groups, and this type of systematic cross-accounting is beyond the reach of the descriptive approach.

One potentially fruitful tactic may be to reinterpret the findings from the descriptive approach using the restrictions implied by factor demand theory. A descriptive national-level regression, such as the one specified in equation (4.7), relates the wage of a skill group to the "own" supply shift, and excludes all the other supply shifters. As we saw in Chapter 4, the wage elasticity implied by this approach was −0.37. A straightforward application of the omitted variable bias formula would suggest that this elasticity is obviously measuring not the own wage effect but an amalgam of the own effect and some combination of the cross-effects that have been excluded from the regression. It is easy to illustrate this insight in the simple case where there are 3 skill groups and we observe market conditions for each skill group for a total of $T$ years.

Let $y_{it}$ give the mean log wage of skill group $i$ ($i = A, B, C$) at time $t$ ($t = 1, \ldots, T$); and $m_{it}$ be the relative supply increase experienced by that group ($m_{it} = M_{it}/N_{it}$). Without loss of generality, assume all variables are measured as deviations from the mean. We can describe the structure of the stacked data with the regression model:

$$\begin{aligned} y_{At} &= \beta\, m_{At} + \varepsilon_{AB}\, m_{Bt} + \varepsilon_{AC}\, m_{Ct} + \varepsilon_{BC} \times 0 + \xi_A, \\ y_{Bt} &= \beta\, m_{Bt} + \varepsilon_{AB}\, m_{At} + \varepsilon_{AC} \times 0 + \varepsilon_{BC}\, m_{Ct} + \xi_B, \\ y_{Ct} &= \beta\, m_{Ct} + \varepsilon_{AB} \times 0 + \varepsilon_{AC}\, m_{At} + \varepsilon_{BC}\, m_{Bt} + \xi_C, \end{aligned} \tag{5.18}$$

where, by construction, the coefficients are factor price elasticities ($\varepsilon_{ij} = d \log w_i / d \log L_j$). The system of seemingly unrelated equations in (5.18) allows for the supply shock to have both own and cross effects, while building in two important restrictions. The first is that the "own effect," which is measured by the parameter $\beta$, is the same for all skill groups. The second is the imposition of a symmetry restriction roughly analogous to that implied by factor demand theory, so that the cross-elasticity of factor price $\varepsilon_{ij} = \varepsilon_{ji}$. This is not precisely the restriction implied by the theory, but I will return to this issue shortly.[22]

It is useful to interpret the descriptive approach as estimating a regression using the stacked data in (5.18), but imposing exclusion restrictions. Specifically, let the variable $z$ denote the first regressor in the system (that is, the first column measuring the own effect). The descriptive regression then infers the wage effect of immigration from the bivariate regression coefficient given by $\hat{\beta} = Cov(z, y) / Var(z)$.[23]

It is well known that $\hat{\beta}$ does not estimate the own wage effect of immigration. An application of the omitted variable bias formula implies that:

$$\begin{aligned} \operatorname{plim} \hat{\beta} = \beta &+ [\varepsilon_{AB}\delta_{BA} + \varepsilon_{AC}\delta_{CA}]\frac{\sigma_A^2}{\sigma_m^2} + [\varepsilon_{AB}\delta_{AB} + \varepsilon_{BC}\delta_{CB}]\frac{\sigma_B^2}{\sigma_m^2} \\ &+ [\varepsilon_{AC}\delta_{AC} + \varepsilon_{BC}\delta_{BC}]\frac{\sigma_C^2}{\sigma_m^2}, \end{aligned} \tag{5.19}$$

where $\delta_{ij} = Cov(m_i, m_j)/Var(m_j)$, the coefficient from an auxiliary regression measuring how an increase in the size of group $j$ changes the size of group $i$; $\sigma_i^2 = \sum_t m_{it}^2 / T$, and $\sigma_m^2 = \sigma_A^2 + \sigma_B^2 + \sigma_C^2$.

Equation (5.19) has an interesting interpretation. Each of the bracketed terms gives the predicted wage impact of the cross-effects associated with

an increase in own supply. For example, the first bracketed term specifically measures the cross-effects associated with an increase in the supply of group *A*. Such an increase is correlated with a particular increase in the size of groups *B* and *C* (as measured by the respective δ's), and each of these supply shifts would have wage effects (as measured by the respective ε's).

The application of the omitted variable bias formula thus indicates that the estimated $\hat{\beta}$ measures an amalgam of the own wage effect plus an "average" of the cross-effects that were excluded from the regression. As a result, the coefficient from the descriptive regression has an economically meaningful interpretation: it is the *total* effect on the wage level resulting from, say, a 10 percent increase in the supply of the own group *and* the associated shift in the supply of all the other groups. Put differently, $\hat{\beta}$ measures the average wage impact of a "generalized" supply shock.

There is one minor caveat. The elegant averaging of cross-effects in equation (5–19) depends crucially on the assumption that factor price cross-elasticities, $\varepsilon_{ij}$, are symmetric. Factor demand theory, however, does not imply this type of symmetry. Instead, factor demand theory implies symmetric cross-elasticities of complementarity, $c_{ij}$. However, the theory also implies that $\varepsilon_{ij} = s_j\, c_{ij}$. This proportionality property suggests a simple way for building in the appropriate symmetry restrictions into the stacked system in equation (5.19): Redefine the supply shift variables as $(s_j \times m_j)$ so that the respective coefficient measures only the elasticity of complementarity.[24]

The "trick" of interpreting the descriptive evidence by using insights from factor demand theory has one crucial advantage over the nested CES framework. The estimate of the average wage effect of immigration is inferred directly from observation and is not built in by functional form assumptions. It is certainly the case that the descriptive approach cannot estimate the structure of cross-effects that would allow us to fully predict the impact of any future supply shock, but it is also the case that currently available structural predictions of the average wage effect are not based on any "real world" observation.